Internship Project Report

INFOSYS SPRINGBOARD

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PROJECT REPORT

ON

AI-Driven Football Player Transfer Value Prediction System

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Abstract

Football transfer values fluctuate dynamically under the influence of multiple intertwined factors such as player performance, age, injury history, market sentiment, league competitiveness, and social popularity. Traditional valuation approaches struggle to grasp these nonlinear complexities. This internship project presents an AI-driven Transfer Value Prediction System that integrates diverse datasets, applies sophisticated preprocessing, and leverages machine learning models—XGBoost and LSTM—to forecast player market values. Delivered via a Streamlit interactive web interface, the system provides fair-market value estimates, analytical insights, and comparative visualisations through radar and bar charts. The resulting solution demonstrates accuracy, scalability, and practical application of AI in sports analytics.

Introduction

The global football transfer market is a vibrant multi-billion-dollar ecosystem where clubs invest heavily in understanding player value to maximise return on investment. However, accurately predicting transfer fees remains challenging due to various nonlinear and hidden factors influencing valuations. This project aims to develop a comprehensive, data-driven automated system that estimates transfer values by considering multiple dimensions, including performance metrics, historical market values, player age and position, sentiment analysis from social media, popularity measures, league competitiveness, and injury frequency. The ultimate goal is to deliver a robust AI pipeline encompassing data collection, cleaning, processing, modelling, and visualisation for reliable transfer value predictions.

Project Objectives

Primary Objectives

- Develop a predictive model to estimate football player transfer values with high accuracy.
- Integrate multi-source heterogeneous data to enhance model robustness.
- Build an end-to-end AI pipeline from data extraction through to deployment.
- Design an intuitive user interface enabling real-time transfer value predictions.

Secondary Objectives

- Conduct feature engineering to derive meaningful player attributes.
- Validate models using exploratory data analysis and performance metrics.
- Visualize analytical patterns such as trends, correlations, and player comparisons.

Data Sources

Data was aggregated from five distinct platforms to form a comprehensive dataset:

Source	Content	Format
GitHub Player Stats	Goals, minutes, assists, expected goals (xG), expected assists (xA)	CSV
TransferMarket (Web Scraping)	Market value history, transfer fees	HTML scraped into CSV
StatsBomb	Detailed performance metrics	JSON
Twitter API	Sentiment and popularity data from tweets	JSON
Online Injury Databases	Player injury history	CSV

The merged dataset comprises approximately 40 columns, containing thousands of rows across multiple seasons for each player, ensuring temporal depth and breadth.

Data Cleaning & Preprocessing

Managing multi-source data posed challenges related to inconsistency and quality. Key cleaning steps included:

- Removing missing or corrupted records to ensure data integrity.
- Standardizing monetary values across currencies (€,,$) into a unified format.
- Eliminating duplicate entries and harmonising textual fields.
- Normalising numerical variables such as age, height, and minutes played for consistency.

Subsequent preprocessing involved:

- One-hot encoding categorical variables to convert them into machinereadable format.
- Scaling numeric features for balanced model training.
- Engineering the target variable `log_value` by applying logarithmic transformation to market values, reducing skew.
- Synchronising datasets per player to create a unified machine learning-ready matrix.

Feature Engineering

To enrich model input, several domain-specific features were crafted as shown below:

Feature	**Description**
Value_per_minute	Market value normalized by minutes played, reflecting efficiency
Injury_frequency_score	Weighted score quantifying injury occurrences and severity
Age_performance_ratio	Performance metrics adjusted relative to player age
Rolling_MV_Avg	Smoothed season-to-season average of market value to capture trends
Sentiment_score & Polarity	Aggregate public sentiment extracted from Twitter data
Popularity_total	Composite social media popularity indicator

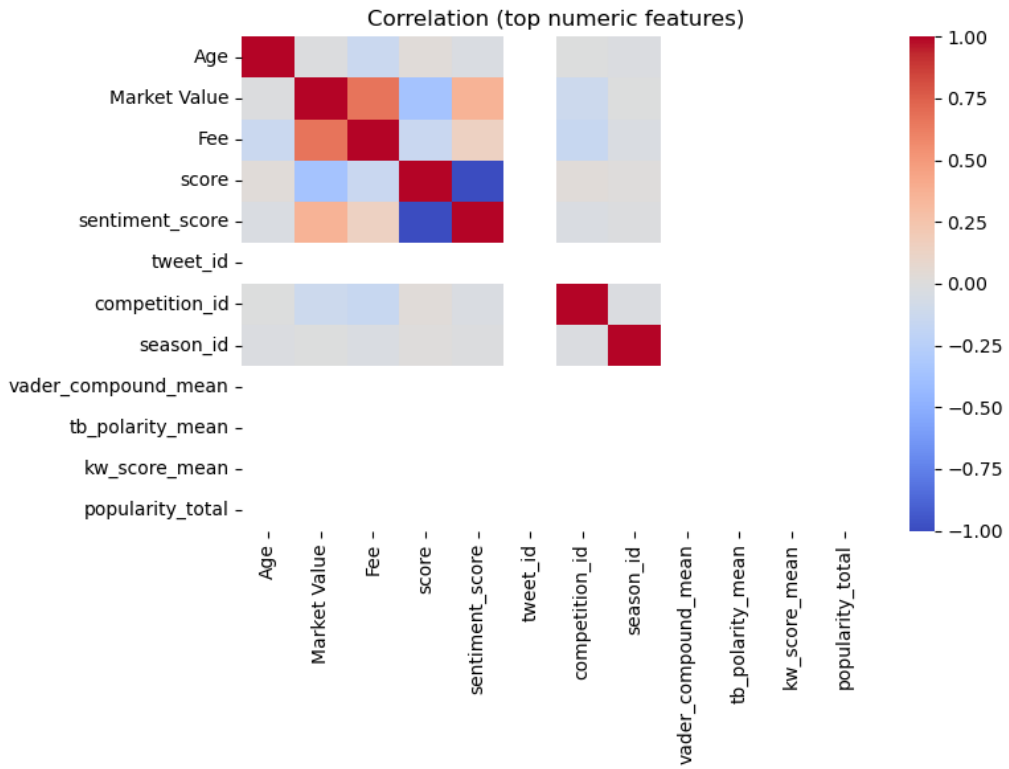
These engineered attributes notably increased the predictive accuracy of the XGBoost model by capturing nuanced player characteristics.

Exploratory Data Analysis (EDA)

Comprehensive EDA provided valuable insights into the dataset structure and relationships:

- Correlation heatmaps revealed strong feature-to-value dependencies guiding feature selection.
- Bar charts illustrated variations in market value across different competitions.
- Scatter plots highlighted the association between player performance and market value.
- Density plots exposed skewness in value distributions, reinforcing the choice of log-transform.

These visualisations informed model design and ensured alignment with underlying data patterns.



Modeling Approach

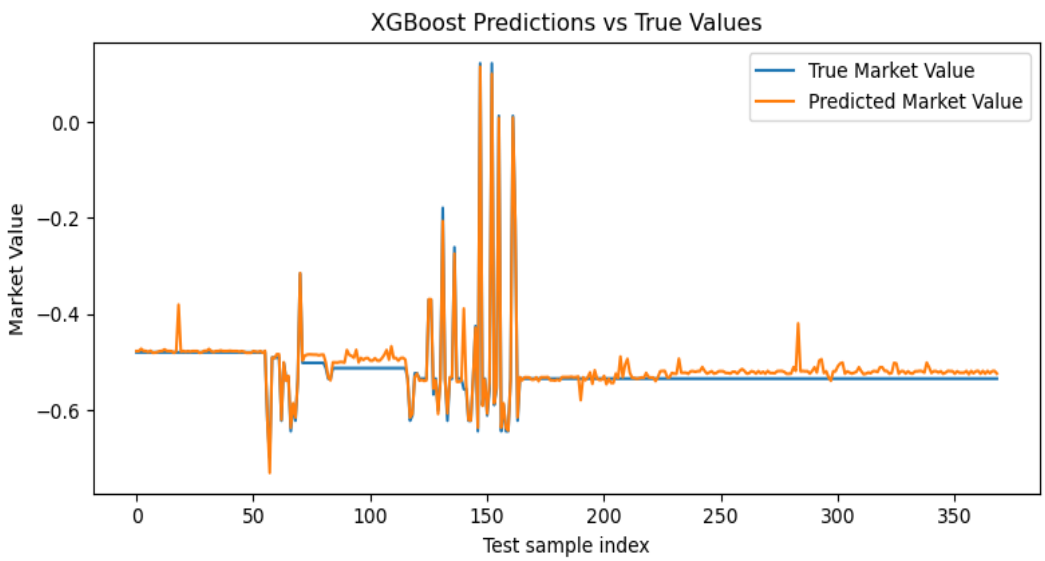
The system employs a dual-model strategy integrating both machine learning and deep learning techniques to leverage their complementary strengths.

XGBoost Regression Model

XGBoost was selected for its robustness on structured tabular data and ability to model complex nonlinearities.

- **Inputs:** Engineered numerical features, sentiment scores, and performance indicators.
- **Output:** Log-scaled predicted market value (`log_value`).
- **Hyperparameters:**
  - `n_estimators` = 500
  - `learning_rate` = 0.05
  - `max_depth` = 6
  - `subsample` = 0.8
  - `colsample_bytree` = 0.8
- **Benefits:** Handles missing data, fast training, strong performance on tabular datasets.

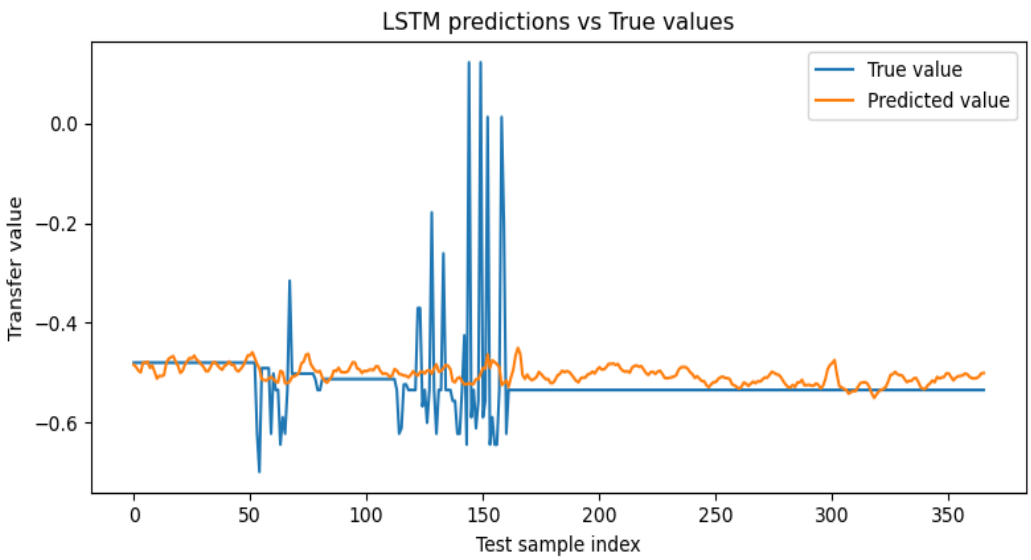
XGBoost consistently yielded the lowest root mean square error (RMSE) among tested models.



LSTM Model for Time-Series Trend Forecasting

To exploit temporal dynamics in market value history, a Long Short-Term Memory (LSTM) recurrent neural network was employed.

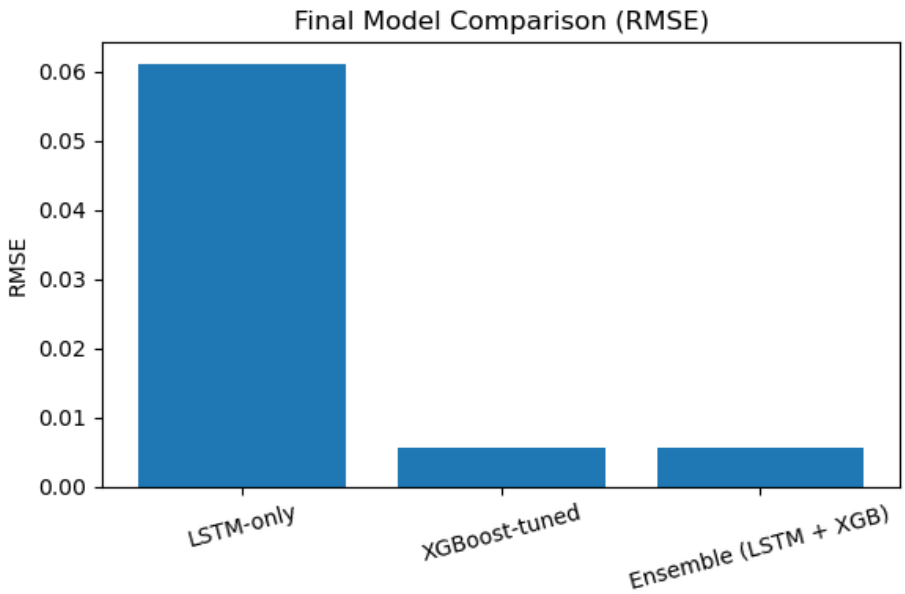
- **Rationale:** LST effectively captures sequential dependencies and progression trends over seasons.
- **Architecture:** Input layer feeding one or more LSTM layers followed by dense regression output.
- **Training:** Utilised Adam optimiser with mean squared error loss.
- **Insights:** Best suited for players with extensive historical data, primarily supplements trend forecasting rather than absolute valuation.



Ensemble Interpretation

The final prediction system combines:

- **XGBoost** as the core estimator for absolute value prediction.
- **LSTM** serving as an optional historical trend predictor, augmenting interpretability



System Architecture

The overall pipeline is designed for modularity and scalability:

Pipeline Stages
1. Data Extraction from multiple sources
2. Cleaning & Preprocessing to unify and standardise data
3. Feature Engineering to derive meaningful variables
4. Model Training: XGBoost and LSTM models
5. Integration Layer combining model outputs
6. Deployment via Streamlit interactive UI

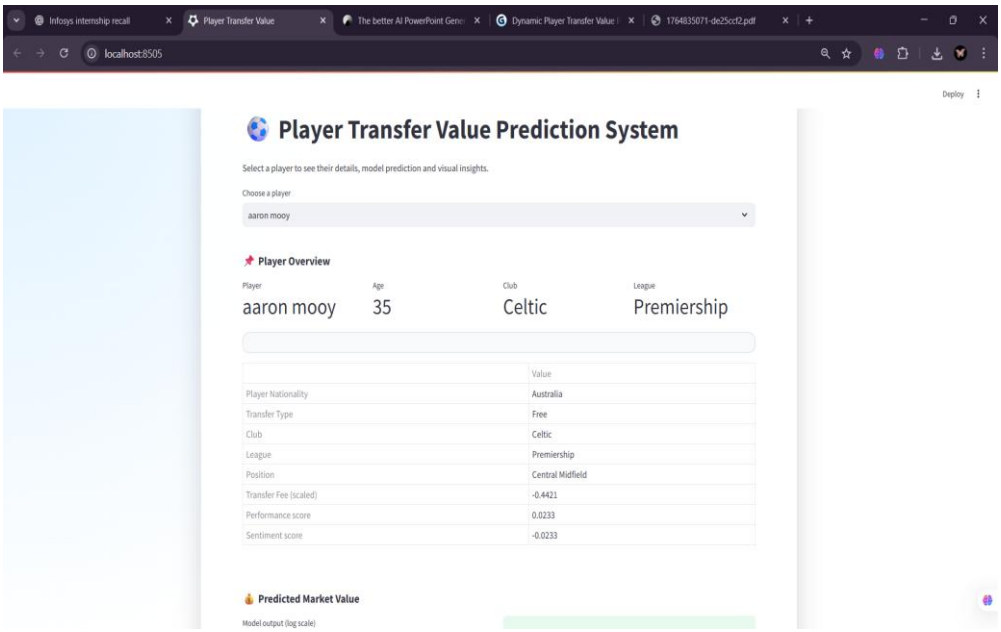
Streamlit Application

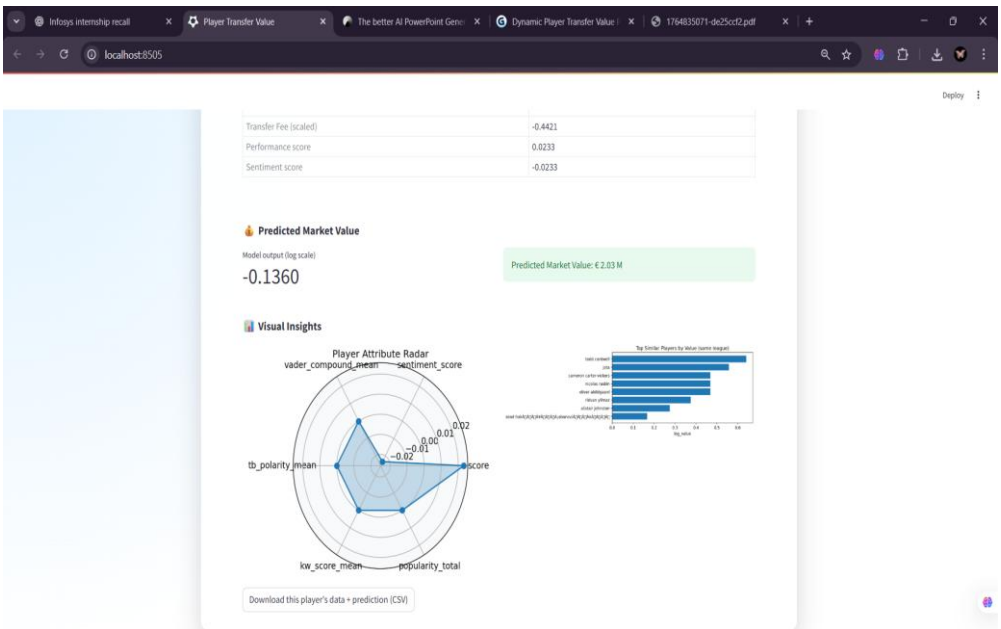
A user-friendly web interface was developed to facilitate real-time interaction with the prediction system.

Key Features

- Dropdown menu for player selection with automatic age adjustment.

- Display of player overview, including club, league, and position.
- Radar charts visualising player attributes.
- Comparison graphs highlighting similar players.
- Predicted transfer value formatting with currency symbols.
- Option to export prediction results as CSV files.





User Flow

Upon selecting a player, the system loads corresponding data and executes model predictions. Visualizations update dynamically to reflect insights. Feedback from mentors highlighted the UI as clean, intuitive, and effective.

Results & Evaluation

Model performance was assessed using multiple metrics:

Metric	**Description**
Mean Absolute Error (MAE)	Average absolute prediction error
Root Mean Square Error (RMSE)	Square root of average squared errors
R^2 Score	Proportion of variance explained by the model

Key Findings

- XGBoost achieved the lowest RMSE, demonstrating superior accuracy.
- LSTM captured short-term market fluctuations but was less precise in absolute valuation.
- The ensemble approach yielded stable and logically consistent predictions.
- Visual plots comparing true versus predicted values confirmed reliability and robustness.

Challenges & Solutions

Challenge	**Solution**
Unstructured monetary formats across data sources	Developed a custom currency parser to unify values
Missing values in multiple datasets	Applied imputation techniques combined with intelligent merging logic

Sparse player historical data limiting sequence models	Adopted a hybrid modelling approach leveraging XGBoost and LSTM selectively
Noisy sentiment data from tweets	Implemented token cleaning and polarity scoring to enhance signal quality
Age provided as z-scores in some sources	Designed reverse scaling logic to recover actual age values

Conclusion

This internship project successfully developed an AI-driven football player transfer value prediction system by integrating multi-source data, advanced preprocessing, and cutting-edge machine learning techniques. The final solution delivers accurate valuation predictions supported by insightful visualisations and a deployable web interface, illustrating practical AI applications in sports analytics.

Future Scope

Potential directions for extending this work include:

- Implementing a real-time web scraping pipeline for up-to-date data ingestion.
- Adopting transformer-based models for enhanced sentiment analysis.
- Incorporating injury severity scoring for deeper health impact analysis.
- Deploying the system on cloud platforms such as AWS or GCP for scalability.
- Adding a recommendation engine to assist scouting decisions.

References

- Pandas Documentation: https://pandas.pydata.org/docs/—Utilised for data cleaning, manipulation, and preprocessing.
- NumPy Documentation: https://numpy.org/doc/—Employed for numerical computations and feature calculations.

- Scikit-Learn Documentation: https://scikit-learn.org/stable/documentation.html.html—Guided model training workflows, evaluation metrics, and preprocessing methods.
- XGBoost Documentation: https://xgboost.readthedocs.io/—Reference for training the regression model on player data.
- Streamlit Documentation: https://docs.streamlit.io/—Used to build the interactive user interface.
- Python Official Documentation: https://docs.python.org/3/—Consulted for core language features and libraries.